

Destructive impact of imperfect beam collimation in extraordinary optical transmission

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We investigate the difference between analytic predictions, numerical simulations, and experiments measuring the transmission of energy through subwavelength, periodically arranged holes in a metal film. At normal incidence, theory predicts a sharp transmission minimum when the wavelength is equal to the periodicity, and sharp transmission maxima at one or more nearby wavelengths. In experiments, the sharpest maximum from the theory is not observed, while the others appear less sharp. In numerical simulations using commercial electromagnetic field solvers, we find that the sharpest maximum appears and approaches our predictions as the computational resources are increased. To determine possible origins of the destruction of the sharp maximum, we incorporate additional features in our model. Incorporating imperfect conductivity and imperfect periodicity in our model leaves the sharp maximum intact. Imperfect collimation, on the other hand, incorporated into the model causes the destruction of the sharp maximum as happens in experiments. We provide analytic support through an asymptotic calculation for both the existence of the sharp maximum and the destructive impact of imperfect collimation. © 2013 Optical Society of America

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1. INTRODUCTION

The subject of extraordinary transmission (EOT) through subwavelength-sized holes in a metal plate or film is well documented (see [1] for a recent review), especially since the work of Ebbesen *et al.* in 1998 [2]. Numerical and physical experiments have explored many different hole shapes, hole arrangements, and metallic compositions. However, theory and experimental results consistently differ from each other in the number and behavior of transmission maxima. According to simulations, a deep transmission minimum known as the Rayleigh anomaly exists at wavelengths close to the hole period, and it can be accompanied by multiple transmission maxima that appear at slightly larger wavelengths [3–5]. We focus on the case when the film parameters are chosen such that there are two such maxima in the vicinity of the Rayleigh anomaly. The maximum that is closer in frequency to the Rayleigh anomaly is narrower and is not seen in experiments. The maximum that is farther from the Rayleigh anomaly appears to correspond to the maximum seen in experiments, though it is broader and of lower amplitude in the experiments. We demonstrate that the discrepancy is due to imperfect collimation of the incident beam in experiments.

Briefly summarizing some of the experiments, Ebbesen *et al.* [2] observed a single broad EOT peak in a silver film perforated with a periodic array of circular holes, whereas simulations of this experiment clearly demonstrate two transmission maxima, as discussed above (see Figs. 3 and 4 of [6]).

Similarly, other experiments in the visible [7], terahertz [8–10], subterahertz [11], and microwave [12] parts of the spectrum for various hole sizes and shapes also do not observe the sharp maximum. Given that these experiments span a wide range of values for the metal dielectric constant, the destruction of the sharp maximum is likely due to another physical effect whose origin can be motivated by considering the case of a perfectly conducting metal film.

Our approach uses the basic model developed in [3,4] for computing the total transmission of light through a thin, perfectly conducting metallic film with a spatially periodic perforation. We describe the model in Section 2 with details in Appendix A. In Section 3, we provide analytic support for the existence and origin of the narrow maximum by introducing an approximation to the model that neglects all but the most prominent of the waveguide modes [3]. We find numerically that the approximation is excellent near the Rayleigh anomaly in the frequency range examined in this study. The fact that the exterior admittance becomes infinite at these frequencies allows us to solve the approximate model near them by means of asymptotic analysis (calculation in Appendix B). We calculate two resonances, or poles of the scattering matrix in the complex frequency plane with small imaginary parts, which are clearly associated with the two maxima. These resonances are associated with bound surface waves arising from the periodic hole array and are sometimes referred to as spoof plasmon polaritons [1,3,13].

We proceed in Sections 4, 5, and 6 to include three additional features of the basic model, one at a time, that improve the model as an accurate descriptor of the physical experiments. We seek to identify the feature that will match the experimental results, and thereby bring to light the mechanism that destroys the narrower maximum from the observed profile. All waveguide modes are accounted for in these computations (no one-waveguide mode approximation).

In Section 4, we take finite conductivity into account in the basic model, which involves an asymptotic calculation of the propagation constant of each waveguide mode (calculation in Appendix C). We find that, in the microwave and infrared ranges, this contributes very little to degrading the twin maxima for the array geometry and metal dielectric constants considered here. In the visible range, the size and location of the maxima are affected by the finite conductivity where real (as opposed to spoof) plasmon polaritons play a more prominent role, but both maxima are still evident. These results are supported by additional simulations in COMSOL Multiphysics and Microwave Studio, although these simulations require extreme computational resources. Our model proves to be much more efficient than these commercial packages for the computations at hand, which are near resonance and involve corners in the geometry.

In Section 5, we account for the finiteness of the hole array using approximate Bloch theory. We achieve this by forming a periodically repeating supercell in which the array is surrounded by a large amount of space with no holes. As with the case of finite conductivity, we find that this also affects the transmission very little.

For the third and final model improvement, we incorporate in Section 6 imperfect collimation of the incident radiation by including a small range of incident angles. Surprisingly, previous theoretical/numerical studies have not accounted for this effect, with the notable exception of [14], and the experimental papers do not specify the degree of collimation of the radiation impinging on the film. We use a rigorous averaging technique to determine the total transmission due to the interfering incident fields. We find that this causes both quantitative and qualitative changes in transmission that are consistent with experimental results. We conclude with an analytic argument that demonstrates the destructive effect of imperfect incident beam collimation on narrow maxima, establishing it as the cause of the discrepancy between simulation and experiment.

2. BASIC MODEL

We draw from the work of Martín-Moreno and co-workers [3,4], although other but related methods are available (see, for example, [5,15–20]). We consider an array of holes that is periodic in the x and y directions in a perfectly conducting metal film of thickness h in the z direction. (See Fig. 1. The same formalism may be used to evaluate the case of an array of infinitely long slits, arranged periodically, parallel to each other, as we do in Section 4.)

The core modeling technique is mode matching; the electromagnetic (EM) fields are represented in the exterior of the film with a Rayleigh wave expansion (both propagating and evanescent waves), and inside the holes by a series of waveguide modes. Appropriate matching conditions are then applied on the boundary between the two regions. This

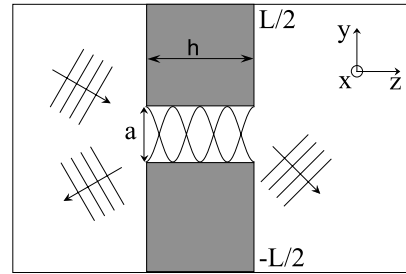


Fig. 1. Plane wave is incident on a metal film of thickness h , with holes of width (in the case of rectangular holes) or radius (in the case of circular holes) a periodically placed with period L . Some of the incident energy is reflected, and some enters the hole, exiting through to the other side.

method has several advantages over more traditional numerical techniques. First, in many cases, only a few waveguide modes contribute significantly to the EM field, which greatly reduces calculation time. Second, decomposing the field into individual modes highlights the emergence of the Rayleigh anomaly. Finally, corners in the metallic sheet and resonances in the transmission cause no significant extra difficulty for the mode-matching technique, while we find that, as expected, COMSOL and Microwave Studio require extraordinary amounts of time and memory to resolve the fields in these circumstances.

The basic model is described in full in Appendix A. The matching process on the two planes marking the boundary of the film reduces to an infinite matrix equation that asserts the continuity of the magnetic field amplitudes across the separating boundary. The unknowns of the equation are the electric field amplitudes of the waveguide modes.

For the perfect conductivity case, periodicity causes the appearance of the Rayleigh anomaly, a deep transmission minimum that occurs when one of the Rayleigh waves outside the film becomes parallel to the surface of the film—an evanescent bound surface wave sometimes called a spoof plasmon polariton [13]. In this condition, the admittance of the p -polarized wave becomes infinite. Under the approximate calculation that uses only a single waveguide mode in the holes, the transmission minimum may take the value zero.

An example of total transmission as a function of wavelength produced by the numerical evaluation of the basic model is shown in Fig. 2 for the case of normal incidence with both square and circular holes. The minimum located at $\lambda/L = 1$ is accompanied by two transmission maxima; the maximum closer to the $\lambda/L = 1$ is narrower than the maximum further away. The next minimum at $\lambda/L = (1/2)^{1/2}$ also has two associated transmission maxima, though they are not so pronounced. The minima are easily located via the frequency condition for the Rayleigh anomaly, but the number, the precise locations, and the heights of the maxima are not so easily predicted and must be found numerically in general. We primarily focus our attention on the twin maxima near $\lambda = L$, and the behavior of the narrower peak in particular, which is general not visible in experimental data as discussed above. Before we continue with our analysis of why the narrower peak is typically not observed in experimental data, we demonstrate analytically the emergence of both maxima by means of an approximation technique.

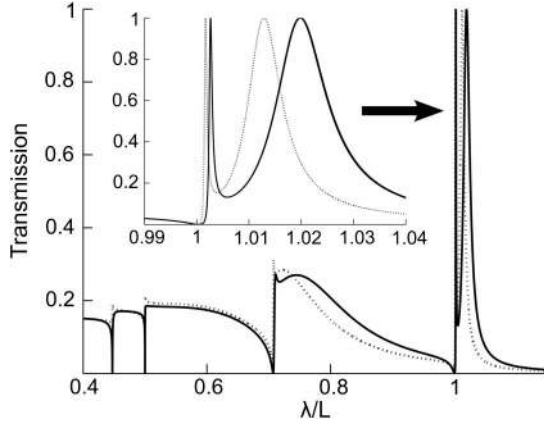


Fig. 2. Transmission for a normally incident plane wave on a two-dimensionally periodic array of square holes. The period $L = L_x = L_y$ is arbitrary, and the thickness of the metal is $h = 0.2L$. The solid line depicts transmission for square holes with side length $a = 0.4L$, and the dotted line depicts transmission for circular holes of the same area. The number λ is the wavelength $\lambda = 2\pi c/\omega$. For both square and circular holes, there are several minima present at $\lambda/L = 1$, at $\lambda/L = (1/2)^{1/2}$, etc. Inset: closeup of the double maximum.

3. FREQUENCY-LOCALIZED APPROXIMATION USING ONE WAVEGUIDE MODE

We find that the field outside the film couples primarily to only a single waveguide mode in the square-hole case shown in Fig. 2. We therefore introduce an approximation of neglecting all but the single mode in our model equations, which produces the same transmission profile to nearly within the numerical accuracy. This is the case for normal incidence, as well for a finite angle of incidence that is p -polarized along either the x or the y direction. Within this approximation, we calculate analytically the resonances at frequencies that are very consistent with those of the transmission maxima. The asymptotic nature of this calculation suggests that it could be performed without the single-mode approximation at the cost of possibly considerable complication.

We outline the main features of the calculation here, while the details of the calculation are presented in Appendix B. The unique allowed mode may exist in two directions along the hole axis so that the final matrix equation of the basic model reduces to two scalar equations. The unknowns in the equations are the total electric field amplitudes X and X' on the two squares that mark the ends of the hole, respectively, which are solutions to

$$\begin{aligned} (G - S)X - G^V X' &= I, \\ G^V X - (G - S)X' &= 0, \end{aligned} \quad (1)$$

where the scalars G , G^V , and S and the incident I correspond to infinite matrices in the full basic model. The matrix G , in the basic model, acts on the waveguide mode amplitude of the electric field on each end of a hole to produce the waveguide mode amplitude of the magnetic field on the same end, assuming that the exterior field on this side is outgoing. Thus G is the exterior mode impedance matrix, transformed into a waveguide mode frame. A similar role is played by the combined action of scalars G^V and S on the waveguide mode; the passage from the electric to the magnetic mode amplitude is

effected not by a connection with the exterior, but by a connection to the amplitudes at the other hole end through the field within the hole.

The scalar G plays a special role in Eqs. (1). It is a series involving all the exterior modes as they connect to the single mode in the hole, and thus it must be summed numerically. Here, G incorporates explicitly the exterior wave admittances Y_k for the plane waves enumerated by k , which exhibit rapid variation, tending to infinity for the p -polarized variety and zero for the s -polarized variety as the frequency approaches that of the Rayleigh anomaly:

$$\begin{aligned} iY_{k,p} &= \frac{\omega/c}{\left(\left(\frac{2\pi}{L}\right)^2 - \left(\frac{\omega}{c}\right)^2\right)^{1/2}} = \frac{1}{\left(\left(\frac{\lambda}{L} - 1\right)\left(\frac{\lambda}{L} + 1\right)\right)^{1/2}}, \\ -iY_{k,s} &= \frac{\left(\left(\frac{2\pi}{L}\right)^2 - \left(\frac{\omega}{c}\right)^2\right)^{1/2}}{\omega/c} = \left(\left(\frac{\lambda}{L} - 1\right)\left(\frac{\lambda}{L} + 1\right)\right)^{1/2}, \end{aligned} \quad (2)$$

where the wavelength is $\lambda = 2\pi c/\omega$. The wavelength parameter λ defined in this way is an analytic parameter that may take complex values. The remaining quantities vary smoothly near the wavelength $\lambda = L$. Therefore we make the further simplification of replacing each of these smoothly varying quantities with their constant, linear, or higher-order expansions centered at the wavelength $\lambda = L$.

In Fig. 3, we show the transmission profile calculated using this method. In the solid line, we recreate the profile from Fig. 2 using one waveguide mode in the holes. As seen in Fig. 3 of [3], this changes the transmission profile to a small but noticeable degree, shifting the broader transmission peak to higher wavelengths. The reason for this deviation is that we neglect the weak coupling of the incident to the majority of the waveguide modes, retaining only the single waveguide mode, which couples strongly to the incident. In the dotted

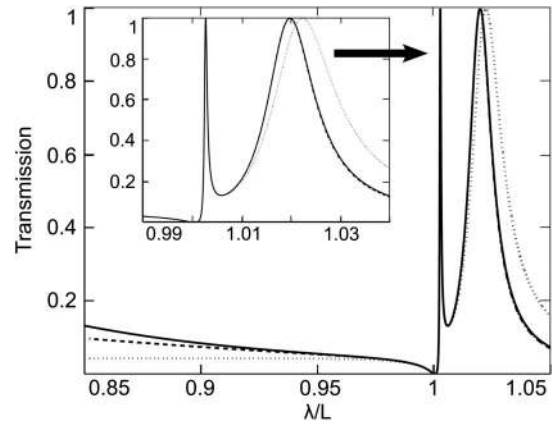


Fig. 3. Transmission profile of a square-hole array using different approximation methods. The solid profile uses the same physical parameters as in Fig. 2, except that only one waveguide mode is included in the calculations. This has the effect of shifting the broad maximum to higher wavelengths, as seen in Fig. 3 of [3], but this shift is slight. The dotted profile, in addition, replaces all nonsingular terms from Eq. (1) with their constant values at $\lambda = L$, finding that this simplification shifts the broad maximum noticeably more to the right. In the dashed line we instead replace nonsingular terms with their linear expansions about $\lambda = L$ and find that the locations of both peaks match those calculated using only the single-mode simplification. Between $\lambda = 0.95L$ and $\lambda = 1.05L$, the dashed and solid curves are practically indistinguishable. Inset: closer view of the peaks.

Table 1. Transmission Maxima Compared to Approximate Values of Resonances λ_{left} and λ_{right} Using Only One Waveguide Mode with Increasing Degrees of Accuracy

	λ_{left}/L	λ_{right}/L
Up to δ^2	$1.002647 + 0.00034i$	$1.022413 + 0.00877i$
Up to δ^3	$1.002623 + 0.00033i$	$1.020092 + 0.00640i$
Up to δ^4	$1.002623 + 0.00033i$	$1.020078 + 0.00638i$
Maxima	1.002668	1.01989

line, we replace smoothly varying quantities with their constant values at $\lambda = L$, finding that the location of the broad maximum is shifted slightly more to the right, although the thin maximum is approximated almost perfectly. Finally, in the dashed line, we replace smoothly varying quantities with their linear expansions about $\lambda = L$ and find that both maxima from the single-mode calculation are approximated to nearly numerical precision.

The two transmission maxima are due to two resonances that occur in the complex frequency plane, slightly off the real axis, and correspondingly at λ_{left} and λ_{right} in the complex λ plane. In an analytic calculation given in Appendix B, we find the locations of these resonances as solutions of the zero determinant condition

$$(G - S)^2 - (G^V)^2 = 0. \quad (3)$$

To include the square root singularities of the admittances in a proper way, we expand the equation for the resonances in powers of $\delta = (\lambda/L - 1)^{1/2}$. The locations of the resonances obtained by retaining terms up to δ^2 , δ^3 , and δ^4 in this equation are given in Table 1. Notice that the convergence is fast as the degree of the approximation is increased. The real parts of the resonances are very close to the wavelengths of the transmission maxima calculated from the full single-mode simulation. The close linkage of these phenomena to the Rayleigh frequency is demonstrated by the fact that the EOT and the resonances that are responsible for it are obtained to great accuracy through expansions about this frequency. Such linkage is not always seen in EOT. For example, EOT is observed at a frequency that is far from the Wood anomaly in a dielectric structure [21].

In each of the next three sections, we seek to identify an improvement to the model that qualitatively matches the mechanism that obscures the narrow maximum.

4. FINITE CONDUCTIVITY

Many commonly used metals have high conductivity and behave similarly to a perfect electric conductor (PEC). However, the calculation of EM fields surrounding a finite conductor is significantly more complicated and requires some approximation. One method is discussed, for example, in [22,23]. Here, we propose a method for the case of large but finite conductivity that is exact except for one asymptotic calculation. We note that our general approach is an approximate solution to Maxwell's equations that accounts for propagating, evanescent, and bound surface waves (surface plasmon polaritons).

Our mode-matching method requires explicit calculation of the waveguide modes inside the holes, which limits the hole

shapes that we are able to consider. We use circular holes for the case of finite conductivity. The reason for this is that we know the explicit form of the waveguide modes in a circular hole given, for example, in [24,25]. (If we did not have these, significantly more numerical calculation would be necessary.) However, two details remain. The first detail is the overlap of waveguide modes between neighboring holes. In a good conductor, however, this feature is negligible, because the field decays quickly and exponentially inside the metal. The second, more pressing, detail is the calculation of the propagation constant k_a of the waveguide modes.

For a real dielectric constant ϵ_1 inside the holes and a real dielectric constant ϵ_2 in the perforated thin film, it is known that the propagation constants of the waveguide modes can be bounded and ordered [24,25], but for complex ϵ_2 the propagation constant is also complex. The propagation constant is the solution to the equation [24]

$$\begin{aligned} & \left(\frac{J'_n(a\lambda_1)}{a\lambda_1 J_n(a\lambda_1)} + \frac{K'_n(a\lambda_2)}{a\lambda_2 K_n(a\lambda_2)} \right) \\ & \left(\omega^2 \mu \epsilon_1 \frac{J'_n(a\lambda_1)}{a\lambda_1 J_n(a\lambda_1)} + \omega^2 \mu \epsilon_2 \frac{K'_n(a\lambda_2)}{a\lambda_2 K_n(a\lambda_2)} \right) \\ & = n^2 k_a^2 \left(\frac{1}{(a\lambda_1)^2} + \frac{1}{(a\lambda_2)^2} \right), \end{aligned} \quad (4)$$

where J_n and K_n are the Bessel functions, a is the radius of the circular hole, $\lambda_1^2 = \omega^2 \mu \epsilon_1 - k_a^2$, $\lambda_2^2 = k_a^2 - \omega^2 \mu \epsilon_2$, and the magnetic constant is assumed to be μ everywhere. For the film geometry considered here, k_0 is complex so that the waveguide mode is evanescent.

Equation (4) cannot be solved for k_z analytically, and it poses numerical challenges as well. It may, however, be solved asymptotically. A PEC is characterized by the dielectric constant $|\epsilon| = \infty$. More precisely, real metals have dielectric constants approaching $\epsilon = i\infty$ in the frequency range in question. By taking $1/\epsilon_2$ approaching zero and the propagation constant k_a approaching that of the associated waveguide mode in a PEC k_0 , we find that

$$k_a \sim k_0 + \frac{C_1}{\epsilon_2^{1/2}} + \frac{C_2}{\epsilon_2} \quad (5)$$

to order $1/\epsilon_2$, where C_1 , C_2 are complicated constants described in Appendix C.

The derivation of this asymptotic expansion is too complicated to produce analytic error estimates. To partially confirm the accuracy of these few terms, we calculate transmission in the microwave, infrared, and visible wavelength ranges twice: once with only the first-order correction term and again with the second-order term. In all three wavelength ranges the two profiles are indistinguishable.

Once we have explicit expressions for the modes, the rest of the mode-matching procedure requires only slight modification, as described in Appendix A. Sample transmission profiles using the same parameters as Fig. 2 are shown in Fig. 4 for three different values of ϵ_2 , in the microwave, infrared, and visible ranges. The deep transmission minimum is present at nearly the same location as it was previously. The profile in the microwave range overlaps everywhere with the profile produced by the PEC, and so the latter is not shown. For

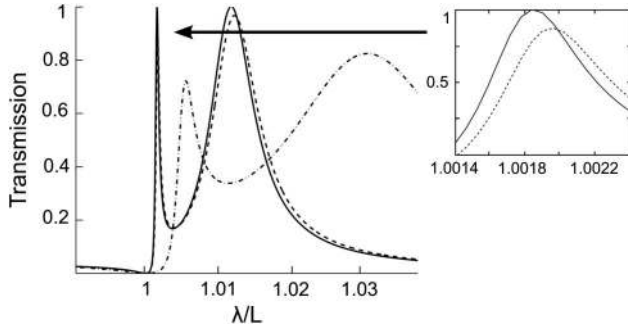


Fig. 4. Transmission profiles using the same parameters as Fig. 2 for the case of finite conductivity. The solid line is for $\epsilon_2 = -7 \cdot 10^4 + i5 \cdot 10^7$, a typical value of the dielectric constant of copper in the microwave range [26]. The values for Al, Cu, and Ag do not differ enough to produce visible change in the transmission profile. It is also indistinguishable from the profile produced by circular holes in a PEC, as in Fig. 2. (In reality, the value of ϵ_2 changes with λ , but again this dependence does not produce any visible difference.) The dashed line is for $\epsilon_2 = -1.7 \times 10^4 + i2 \times 10^4$, a typical value of the dielectric constant of silver in the infrared range. The primary difference is that the height of the narrow peak has decreased to near 90%. Finally, the dash-dotted line is for $\epsilon_2 = -40 + i12$, a typical value of the dielectric constant of aluminum in the visible range. Inset: closeup of the narrow maximum.

the profile produced in the infrared range, the maxima are slightly lower, while for the profile produced in the visible range, the maxima are significantly lowered and shifted to higher wavelengths, but are still present and clear. For other film parameters (especially considering film thickness), surface plasmon polaritons that occur in finite-conductivity films can have [14] or not have [22,23] a destructive effect on the sharp transmission maximum.

We also undertake the same simulation numerically in COMSOL Multiphysics, a finite-element-based commercial EM software. The geometry of the model used in COMSOL is shown in Fig. 5(a): a plane wave is incident from one boundary plane, and the transmitted energy is collected on the other. The boundary conditions are periodic, since incident energy is normal to the film. Two different films are simulated: a PEC, and Al at frequency 10 GHz. In our initial simulations (not shown), we use a default finite-element mesh, but find that the sharp transmission maximum is barely visible. Increasing the mesh density near the corners of the holes results in predictions that approach closer and closer to

our modal-analysis results. Figure 5(b) is for the case in which we use the full resources allowed by our computer cluster, where it is seen that the numerical simulation does not reproduce the unit height of the narrow transmission maximum, although it does reproduce the broad maximum. It does, however, confirm that the perfectly conducting metal and the Al film exhibit indistinguishable transmission profiles.

In greater detail, the disagreement between the numerics and our theory is due to the finite mesh and imperfect absorption at the radiation boundary plane in the simulator. A denser mesh helps the numerical results to match better with the theoretical prediction at the cost of more computing resources. The mesh used to create Fig. 5 has more than 170,000 mesh elements and creates more than 1 million degrees of freedom, which requires more than 32 GB of memory. Second, the imperfect absorption boundary properties at the incident and evaluation planes generate extra oscillations not pictured here.

We also simulate numerically the transmission profile in Microwave Studio, in which we have better control of the reflection at the boundary. Microwave studio, also known as CST, is a commercial numerical EM solver based on the finite-difference time-domain methods. It belongs to the general class of the grid-based differential time-domain numerical modeling method, which is different from the finite-element method in COMSOL Multiphysics. In Microwave Studio, we specify the absorption rate at the excitation and evaluation boundary to minimize numerical reflection errors, a feature that is not supported in COMSOL. However, this approach does affect the predicted transmission curves. The results are shown in Fig. 6. Nearly all features of the profile produced by the modal technique are reproduced, but compared to the modal method and COMSOL, Microwave Studio overestimates the transmission almost everywhere. The exception to this is the narrow transmission maximum, which is present but underestimated compared to both other methods. As with the COMSOL simulation, increasing the calculation density causes the transmission profile to become closer to the one created by the modal method.

5. INCORPORATION OF THE FINITENESS OF THE EXPERIMENTAL ARRAY

Our mode-matching method inherently relies on periodicity, but we can simulate the behavior of a finite hole in the

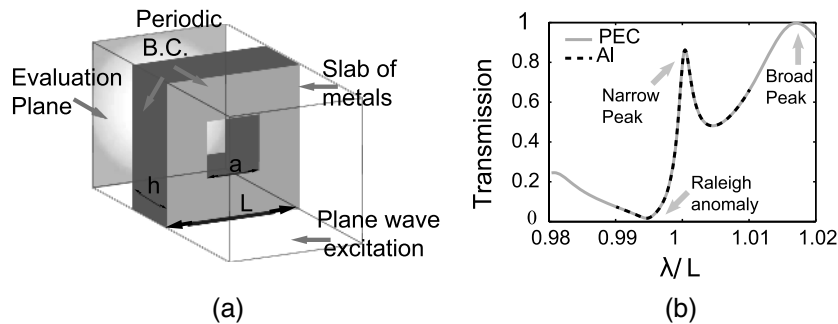


Fig. 5. Transmission profile calculated using the same parameters Fig. 4 produced by COMSOL Multiphysics instead of our model. (a) A plane wave is incident from one boundary, and transmitted energy is calculated at the other. Because we assume normal incidence, periodic boundary conditions are enforced on the four sides of the period cell. (b) Two different films are simulated, one that is a perfect electric conductor (PEC), and another that is Al at a frequency of 10 GHz. The transmission profiles of both metals agree. The Rayleigh anomaly and the narrower transmission maxima, however, are not captured exactly. The transmission profile becomes closer to the one simulated by the modal method as the number of mesh points increases.

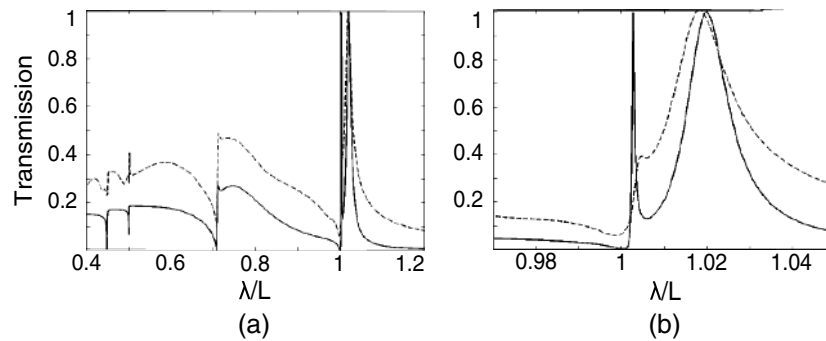


Fig. 6. Transmission calculated using the same parameters as Fig. 4 produced in Microwave Studio. In both pictures, the solid line is the true transmission profile, and the dashed line is the profile produced by Microwave Studio. (a) Compared to our simulations and the COMSOL results, the transmission is overestimated almost everywhere. (b) Closeup of the double maximum. The narrow maximum is only hinted at, and the broad maximum is more closely captured. As with the COMSOL simulations, in the range tested, increasing the calculation density causes the transmission profile to become closer to the one simulated by the modal method. However, it is too computationally expensive to increase the density further.

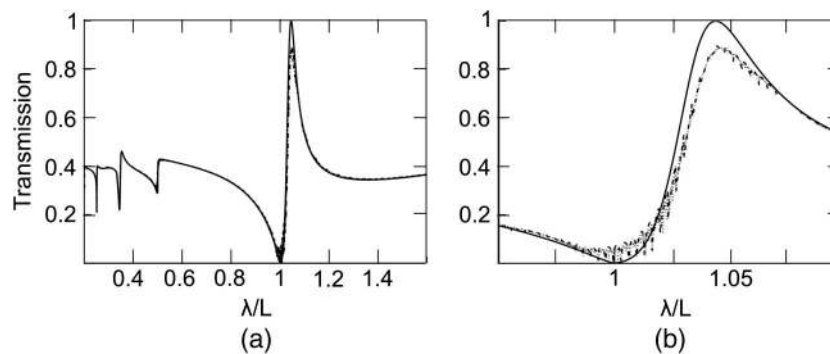


Fig. 7. (a) Transmission profile for a plane wave normally incident on a one-dimensionally periodic array of slits in a perfectly conducting film, with distance L between neighboring holes, width $0.4L$, and film width $0.2L$. (b) Closeup comparison of the previous profile (solid line) near $\lambda/L = 1$ with the transmission profile produced from a finite array of 30 holes repeated periodically with distance $300L$ between arrays, and also with a normally incident plane wave (dashed line). The latter curve is indistinguishable from the curve in (a), and it is also indistinguishable from one produced using distance $1500L$ between arrays with no other change in parameters.

following way. We define a large supercell consisting of a block of holes followed by a span of metal with no holes and repeat this periodically. As the amount of space between blocks tends to infinity, each block of holes will approach the behavior of a finite array. This method is computationally expensive but requires very little change to the theoretical process, as described in Appendix A.

In Fig. 7, we demonstrate this in the case of a one-dimensionally periodic array of infinite slits. We have chosen to demonstrate the one-dimensional case because of the computational expense of the method. We begin by plotting the analogue of Fig. 2 in one dimension of periodicity: the metal has an infinite periodic array of slits of distance L between them, with energy normally incident. The transmission profile is presented in Fig. 7(a). Next, we created finite arrays of 30 holes, repeated periodically with distance $300L$ between them, again with normally incident radiation. Finally, this was repeated with distance $1500L$ between arrays. The latter two transmission profiles are indistinguishable from each other. They are also indistinguishable from the infinitely periodic array, except near the global minimum and maximum. In Fig. 7(b), the transmission profile for the infinite array is compared with the profile for these finite arrays.

As the number of holes in the finite array increases, it behaves more like the infinite array where the transmission increases as the size increases, consistent with experimental

observations [11]. The same behavior is seen in the two-dimensional array, although the expense of the calculations prevents us from providing useful illustration. With only 30 holes in the one-dimensional array, the transmission is nearly identical to the profile produced from the infinite array. Therefore, we find that assuming infinite periodicity in our calculations does not account for the differences between our modal method and experiments.

6. ACCOUNTING FOR IMPERFECT COLLIMATION

In our basic method, the metal film is illuminated on one side by a plane wave, simulating perfect collimation, which is not possible in experiments, however. We can simulate the effects of imperfect collimation by allowing the film to be illuminated by a superposition of plane waves over a small range of incident angles. The inclusion of this effect in our model is simple, as the transmission profile for multiple incident plane waves is the sum of the transmission profiles for each individual incident wave, as described in Appendix A.

We demonstrate the effects of this in the case of square holes in a square lattice. In Fig. 8, we demonstrate the results of averaging transmission profiles for various angles of incidence within three different angle intervals about normal incidence: 0.1° , 0.2° , and 1° . The thinner of the two peaks near $\lambda = L$

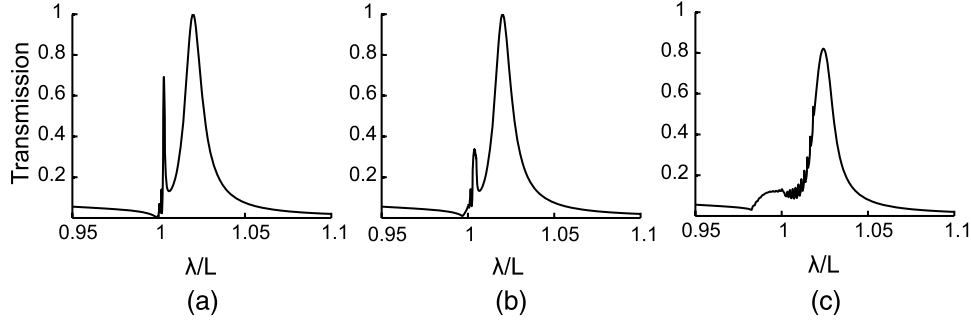


Fig. 8. Transmission profiles for square holes periodically repeated with period L and hole width $a = 0.4L$ in a perfectly conducting film of width $h = 0.2L$, illuminated by plane waves at various angles. (a) Incident waves vary from -0.1° to $+0.1^\circ$ in both the x and y directions. (b) Incident waves vary from -0.2° to $+0.2^\circ$. (c) Incident waves vary from -1° to $+1^\circ$. The thinner of the two transmission maxima diminishes in height until it is no longer apparent, and the single remaining maximum decreases in magnitude more slowly.

diminishes in height until all that remains is a single peak, which also decreases in amplitude but at a slower rate. The transmission minimum remains, but it does not reach as low as it did previously. Our observations are informed by the predictions of Cao and Lalanne [14], who show that the frequency of the sharp transmission maximum occurring near $\lambda = L$ varies rapidly as a function of the incidence angle of the illuminating radiation. We demonstrate here that this rapid angle dependence places tight limits on the degree of collimation of the beam used to illuminate a perforated film in experiments.

7. CONCLUSION

Beginning with a known mode-matching method for evaluating EM scattering from a perforated metal film, we alter the model in several ways. We find that the assumption of perfect conductivity only alters the results minimally in the microwave and infrared ranges, and does not remove the important qualitative behavior even in the visual range. The assumption of infinite periodicity can affect the amount of transmission but does not affect the qualitative shape of the transmission profile, including the transmission minimum and the location of transmission maxima. Allowing the level of collimation to vary, however, substantially affects the transmission profile, causing it to take on the qualitative features of experiments. For the physical parameters considered in this study, the narrow transmission maximum is reduced to 50% height when collimation varies by about 0.15° .

APPENDIX A: THE MODE-MATCHING PROCEDURE

The basic mode-matching procedure may be found in [3,4]. Here we reproduce the resulting formulae for reference. We begin with the case of a perfectly conducting metal, so that the dielectric constant $|\epsilon_2|$ inside the metal is infinite; $\epsilon = 1$ away from the film; ϵ_1 inside the holes is finite, typically 1; and μ is identically 1. The changes made to the procedure in each of the cases discussed in the previous sections will be noted at the end.

The EM field outside the film is written as a superposition of plane waves, enumerated by k and denoted W_k , and the EM field inside the holes is written as a superposition of waveguide modes enumerated by α and denoted M_α . The amplitudes of the electric part of the waveguide modes at the hole openings, denoted by the vector X on the reflecting side

and X' on the transmitting side, are determined by the matrix equation

$$\begin{pmatrix} G - S & -G^V \\ -G^V & G - S \end{pmatrix} \begin{pmatrix} X \\ X' \end{pmatrix} = \begin{pmatrix} I \\ 0 \end{pmatrix}. \quad (\text{A1})$$

The matrix G applied to the electric amplitudes X and X' transforms waveguide mode amplitudes into plane-wave amplitudes using the continuity conditions, then applies the admittances of plane waves to transform electric amplitudes into magnetic, and finally returns the plane-wave amplitudes to waveguide amplitudes. Explicitly, it is

$$G_{\alpha\beta} = i \sum_k Y_k (M_\alpha, W_k) (W_k, M_\beta), \quad (\text{A2})$$

where Y_k is the admittance of plane-wave mode W_k , and the inner products (U, V) with $U = (U_x, U_y, U_z)$ are calculated via the integral

$$(U, V) = \int_{\text{period cell}} U_x^* V_x + U_y^* V_y \, dx dy. \quad (\text{A3})$$

The matrices S and G^V incorporate the effects of reflections from one side of the hole onto the other side. They are given by

$$S_\alpha = i Y_\alpha \frac{e_\alpha^2 + e_\alpha^{-2}}{e_\alpha^2 - e_\alpha^{-2}}, \quad G_\alpha^V = \frac{2i Y_\alpha}{e_\alpha^2 - e_\alpha^{-2}}, \quad (\text{A4})$$

where Y_α is the admittance of waveguide mode M_α , and $e_\alpha = \exp(ik_\alpha h/2)$ for film thickness h . Finally, the vector I denotes the amplitudes of the magnetic components of the incident wave and is given by

$$I_\alpha = 2i Y_{k_0} (W_{k_0}, M_\alpha), \quad (\text{A5})$$

where W_{k_0} denotes the incident plane wave.

Once the amplitudes of the plane waves are known, the amplitudes t_k and r_k of the transmitted and reflected waveguide modes W_k can be recovered with

$$\begin{aligned} t_k &= -\sum_\alpha (W_k, M_\alpha) X'_\alpha, \\ r_k &= -\delta_{k,k_0} + \sum_\alpha (W_k, M_\alpha) X_\alpha. \end{aligned} \quad (\text{A6})$$

The total transmission can then be calculated using

$$T = \sum_k \Re(Y_k) |t_k|^2 / Y_{k_0}. \quad (\text{A7})$$

The plane waves are enumerated by those $k = (k_x, k_y)$ satisfying

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} k_{x_0} \\ k_{y_0} \end{pmatrix} + 2\pi \begin{pmatrix} j_1/L_x \\ j_2/L_y \end{pmatrix} \quad (\text{A8})$$

for all integers j_1, j_2 and incident wavevector (k_{0x}, k_{0y}) . With these values of k , the x and y components of the plane waves W_k above are

$$\begin{aligned} W_{k,p} &= \begin{pmatrix} k_x \\ k_y \end{pmatrix} \frac{e^{i(k_x x + k_y y)}}{\sqrt{L_x L_y} |k_t|}, \\ W_{k,s} &= \begin{pmatrix} -k_y \\ k_x \end{pmatrix} \frac{e^{i(k_x x + k_y y)}}{\sqrt{L_x L_y} |k_t|} \end{aligned} \quad (\text{A9})$$

for p - and s -polarized waves, respectively, and $|k_t| = (k_x^2 + k_y^2)^{1/2}$. The propagation constant of a plane wave is given by $k_z^2 = (\omega/c)^2 - k_x^2 - k_y^2$. The admittance of a p -polarized plane wave is $Y_{k,p} = \omega/(ck_z)$, while the admittance of an s -polarized wave is $Y_{k,s} = ck_z/\omega$.

The form of the waveguide modes M_α depend on the shape of the holes, but they always have z dependence $\exp(ik_\alpha z)$ for some k_α . In the case of a PEC, waveguide modes come in two varieties, transverse electric (TE) and transverse magnetic (TM). The admittance of a TE mode is $Y_{\alpha,\text{TE}} = ck_\alpha/\omega$, and the admittance of a TM mode is $Y_{\alpha,\text{TM}} = \omega/(ck_\alpha)$.

The changes to this method for the case discussed in Section 3 of a local approximation using only one waveguide mode in a square hole are discussed in Appendix B.

For the case of imperfect conductivity discussed in Section 4, we make the following adjustments to the basic model. First, the propagation constants of the waveguide modes cannot be computed exactly, and instead are computed asymptotically. Their values are given in Appendix B. Second, the admittance of the waveguide modes becomes more complicated. Modes no longer divide into TE and TM modes, but are instead hybrid modes that contain elements of each. Thus we require two different admittances for each mode, and each of these also takes different values in the metal and in the hole for a total of four values. If we denote the TE part of the waveguide mode M_α in the holes by $M_{\alpha,\text{hole}}^{\text{TE}}$, and similarly for $M_{\alpha,\text{metal}}^{\text{TE}}$, $M_{\alpha,\text{hole}}^{\text{TM}}$, and $M_{\alpha,\text{metal}}^{\text{TM}}$, then the net result of this effect is a redefinition of the terms S and G^V as

$$S_\alpha = iY^\zeta \frac{e_\alpha^2 + e_\alpha^{-2}}{e_\alpha^2 - e_\alpha^{-2}}, \quad G_\alpha^V = \frac{2iY^\zeta}{e_\alpha^2 - e_\alpha^{-2}}, \quad (\text{A10})$$

where the new combined admittance Y^ζ is equal to

$$\begin{aligned} Y^\zeta &= \frac{\epsilon_1 \omega}{ck_\alpha} (W_k, M_{\alpha,\text{hole}}^{\text{TE}}) + \frac{\epsilon_2 \omega}{ck_\alpha} (W_k, M_{\alpha,\text{metal}}^{\text{TE}}) \\ &+ \frac{ck_\alpha}{\mu \omega} (W_k, M_{\alpha,\text{hole}}^{\text{TM}}) + \frac{ck_\alpha}{\mu \omega} (W_k, M_{\alpha,\text{metal}}^{\text{TM}}). \end{aligned} \quad (\text{A11})$$

Aside from these changes, the matrix equation remains the same.

To approximate a finite array as in Section 5, we make the following adjustments to the basic model. Instead of calculating waveguide modes in only one hole per period cell, we define a larger supercell containing some arrangement of holes, and then repeat this supercell periodically, where we must calculate waveguide modes for each hole in the supercell. Thus, the terms $M_\alpha, Y_\alpha, X_\alpha, X'_\alpha, e_\alpha, G_{\alpha,\beta}, I_\alpha, G_\alpha^V, S_\alpha$ must be enumerated over each hole as well, say $M_{\alpha,l}, Y_{\alpha,l}$, etc. The final matrix equation will accordingly be, in essence, a number of times larger to a side equal to the number of holes in each supercell.

To approximate imperfect collimation as in Section 6, we make the following changes to the basic model. Instead of defining the incident energy to be in the form of a plane wave, we instead define the incident energy to be a superposition of plane waves that differ only slightly in their angles of incidence. Due to linearity of Maxwell's equations and the orthogonality of the plane waves and waveguide modes that led to Eq. (A7), the total transmission in this case is equal to the sum of the transmission caused by each individual incident wave.

APPENDIX B: ONE-MODE APPROXIMATION AND RESONANCE FORMULA

We remarked earlier that, near the Rayleigh frequencies, the approximation of retaining only one waveguide mode in the hole is particularly good for square holes. Moreover, the matrices G, S, G^V and the vectors X, X', I are reduced to scalars in this approximation. Using the notation of [3], we define

$$\Delta = e_\alpha(G - iY_\alpha), \quad \Gamma = e_\alpha^{-1}(G + iY_\alpha), \quad (\text{B1})$$

reducing the formula for total transmission to

$$T = \left| \frac{2iY_\alpha I}{\Delta^2 - \Gamma^2} \right|^2 \frac{\Im(G)}{Y_{k_0}}. \quad (\text{B2})$$

The denominator $\Delta^2 - \Gamma^2$ being zero is equivalent to the zero determinant condition $(G - S)^2 - (G^V)^2 = 0$. This is a resonance condition; an ω satisfying this condition corresponds to a pole of the scattering operator in the one-mode approximation. Setting the denominator equal to zero gives

$$G = iY_\alpha \frac{e_\alpha \pm e_\alpha^{-1}}{e_\alpha \mp e_\alpha^{-1}}. \quad (\text{B3})$$

This condition is not satisfied for real ω ; however, we find two such ω in the complex ω plane very close to the real axis. It is these two resonant values that are responsible for the appearance of the two transmission spikes of Fig. 2 at normal incidence.

Near these two spikes, the most important aspects of Eq. (B3) are those terms from the sum defining G [see Eq. (A2)], which are singular at $\omega = 2\pi c/L$. We split up the expansion of G to the terms with singular admittance Y_k at $\omega = 2\pi c/L$ and the ones that vary smoothly,

$$\begin{aligned} G &= G_c + \frac{i\omega/c}{((\omega/c)^2 - (2\pi/L)^2)^{1/2}} G_p \\ &+ \frac{i((\omega/c)^2 - (2\pi/L)^2)^{1/2}}{\omega/c} G_s. \end{aligned} \quad (\text{B4})$$

In this formula, G_p and G_s consist of the inner products of the waveguide mode with the corresponding p - and s -polarized plane waves, respectively, and are independent of ω , while G_c is the rest of the sum defining G . Inserting this expression into Eq. (B3), dividing by the coefficient of G_p , and making the substitution $\omega = 2\pi c/\lambda$ gives

$$G_p - G_s \left(\frac{\lambda}{L} - 1 \right) \left(\frac{\lambda}{L} + 1 \right) + \left(\left(\frac{\lambda}{L} - 1 \right) \left(\frac{\lambda}{L} + 1 \right) \right)^{1/2} \times \left(G_c - iY_a \frac{e_a \pm e_a^{-1}}{e_a \mp e_a^{-1}} \right) = 0. \quad (\text{B5})$$

We then set $\lambda/L - 1 = \delta^2$, replace the square root with its expansion in powers of δ , and replace the rightmost term in parentheses with its series expansion in δ^2 (since it consists only of terms nonsingular in λ):

$$G_p - G_s \delta^2 (\delta^2 + 2) + \delta \sqrt{2} \left(1 + \frac{\delta^2}{4} + O(\delta^4) \right) \times (\gamma_0^\pm + \gamma_1^\pm \delta^2 + O(\delta^4)) = 0, \quad (\text{B6})$$

where the superscript $\pm$ refers to the choice of positive or negative from Eq. (B3).

By retaining only the constant and δ terms in this equation, we obtain the solution

$$\delta = -\frac{G_p}{\sqrt{2}\gamma_0^\pm}, \quad (\text{B7})$$

where the choice of γ_0^+ corresponds to the peak closer to $\lambda = L$ and the choice of γ_0^- corresponds to the peak farther from $\lambda = L$. Retaining higher powers of δ makes corrections to these solutions; corrections beyond δ^4 are truly negligible (see Table 1).

APPENDIX C: ASYMPTOTIC EXPRESSION FOR THE PROPAGATION CONSTANT

Equation (4) determines the propagation constant k_a for a hybrid waveguide mode in a circular waveguide encased by another material, each of which have finite conductivity ϵ_1 and ϵ_2 , respectively. The magnetic constant μ is taken to be the same in both materials.

If the surrounding material has dielectric constant ϵ_2 approaching infinity, then the hybrid mode will approach a TE or TM mode as if the waveguide were surrounded by a PEC. Thus, we can solve for k_a asymptotically by allowing k_a to be close to k_0 (the propagation constant for the PEC) and letting $1/\epsilon_2$ be close to zero. The first terms of this asymptotic expansion are

$$k_a \sim k_0 + \frac{C_1}{\epsilon_2^{1/2}} + \frac{C_2}{\epsilon_2}. \quad (\text{C1})$$

The constants C_1 and C_2 differ depending on whether the hybrid mode approaches a TE mode or a TM mode. In the TE case, k_0 is such that $u_{p'} = a(\omega^2\mu\epsilon_1 - k_0^2)^{1/2}$ is the p th zero of the Bessel function derivative J'_n , and

$$C_1^{\text{TE}} = \frac{-J_n(u'_p)\alpha}{a(-\omega^2\mu)^{1/2}J'_n(u'_p)k_0} \left(1 + \frac{n^2k_0^2}{a^2\alpha^2} \right), \quad (\text{C2})$$

$$C_2^{\text{TE}} = \frac{-J_n(u'_p)\alpha}{k_0J'_n(u'_p)} \left(1 + \frac{n^2k_0^2}{a^2\alpha^2} \right) \left[\frac{1}{a^2\omega^2\mu} \left(\frac{1}{2} + \frac{n^2k_0^2}{a^2\alpha^2} \right) + \frac{J_n(u'_p)n^2}{J'_n(u'_p)a^2\alpha} \left(2 + \frac{4k_0^2}{\alpha} \right) \right], \quad (\text{C3})$$

where $\alpha = \omega^2\mu\epsilon_1 - k_0^2 = (u'_p/a)^2$, and p is a positive nonzero integer. Meanwhile, in the TM case, k_0 is such that $u_p = a(\omega^2\mu\epsilon_1 - k_0^2)^{1/2}$ is the p th zero of J_n , and

$$C_1^{\text{TM}} = -\frac{\epsilon_1(-\omega^2\mu)^{1/2}}{ak_0}, \quad (\text{C4})$$

$$C_2^{\text{TM}} = \frac{\epsilon_1\omega^2\mu}{2k_0} \left(\frac{1}{a^2k_0^2} - \frac{1}{a^2\beta} + \frac{J''_n(u_p)}{J'_n(u_p)\alpha(\beta^{1/2})} \right) + \frac{\epsilon_1}{2a^2k_0}, \quad (\text{C5})$$

where $\beta = \omega^2\mu\epsilon_1 - k_0^2 = (u_p/a)^2$.

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